INCH-POUND

MIL-A-85097/16(OS)
5 December 1994

MILITARY ASSOCIATED DETAIL SPECIFICATION

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR, PARACHUTE DEPLOYMENT, MARK 122 MOD 0

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Mk 122 Mod 0 Parachute Deployment Rocket Motor described herein shall consist of this specification and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This document covers the requirements for the Mark 122 Mod 0 Parachute Deployment Rocket Motor (PDRM) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents are in addition to those listed in the basic specification, MIL-A-85097, and form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

MBEU146190	Rocket Motor Mk 122 Mod 0
RU-ATP-9	Acceptance Test Procedure for Parachute Extractor Rocket Motor

(Application for copies of specifications and drawings should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Code 8410P, 101 Strauss Avenue, Indian Head, MD 20640-5035.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Code 8420, 101 Strauss Avenue by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The requirements in this associated detail specification, unless in direct conflict with the basic specification, shall be considered additional, not substitute. PDRMs manufactured to this specification shall comply with the requirements specified in Drawing MBEU 146190, Acceptance Test Procedure RU-ATP-9, associated process specifications, and tables I and II. If there is a direct conflict with the basic specification, this document shall take precedence.

3.2 First article.

3.2.1 Propellant aging. When the loaded motor is tested per MIL-A-85097, it shall meet the ballistic parameters listed in tables I and II. The propellant shall meet the physical and mechanical properties of the manufacturer's propellant specification for a minimum of 84 months from the month of extrusion.

TABLE I. Ballistic performance requirements for Mk 122 Mod 0 Rocket Motors (rocket section phase).

Parameter	165 ± 5°F		-65 ± 5°F	
	Min	Max	Min	Max
Impulse, lb-sec	71	87	71	87
Action time, sec	0.214	0.247	0.239	0.328
Maximum thrust, lb	335	421	274	357
Ignition delay, sec	-	0.010	-	0.010

TABLE II. Ballistic performance requirements for Mk 122 Mod 0 Rocket Motors (mortar phase).

Parameter	165 ± 5°F		-65 ± 5°F	
	Min	Max	Min	Max
Gas ignition delay, msec	-	30	-	30
Initiation pressure, psig	400	600	400	600
Velocity, ft/sec	170	250	170	250

3.3 First article/production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

3.3.1 Propellant. The main propellant charge used in the PDRM shall be KU (produced by Royal Ordnance, plc, Bishopton, Scotland), or a Navy-approved equivalent. The propellant shall conform to RO's internal propellant specification (ref: BS-T-480-S05-20129-1) or a Navy-approved equivalent when tested for MIL-A-85097 data. The propellant manufacturer shall make the results of the physical and mechanical tests available to the rocket motor manufacturer(s) for comparison (see 6.2.2).

3.3.2 Life. Life shall conform to the requirement of MIL-A-85097, with the exception of propellant life which shall be 3 years maximum on sheet stock and 1 year maximum on extruded propellant grains.

3.3.3 Ballistic performance. The PDRM shall meet the performance requirements specified in tables I and II when tested in accordance with 4.2.3.

3.3.4 Radiographic inspection. In addition to the radiographic inspection required by MIL-A-85097, the rocket motor components shall be radiographically inspected prior to static firing in accordance with 4.2.4. The PDRM mortar assembly shall be radiographically inspected in accordance with 4.2.4 after reassembly with the simulation billet and prior to firing.

3.3.5 Resistance. The primary cartridge (Drawing MBEU142830) shall have an insulation resistance of 100 megohms minimum at 500 Vdc when shorted pins-to-case. The bridgewire circuits, pin A-D and B-C, shall have a resistance of 0.90 to 1.30 ohms when tested in accordance with 4.3.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 First article vibration testing. For the AEPS covered by this specification sheet, the following test procedures shall be used in lieu of those specified in the general specification.

a. Prior to vibration testing, the rocket motors shall be temperature conditioned for a minimum of 6 hours and a maximum of 48 hours, one at -65 ± 5°F and the other at 165 ± 5°F. The rocket motors shall be maintained at these conditioning temperatures during vibration tests. A continuous temperature history of the rocket motor shall be kept for all conditioning and test times.

b. If the rocket motor is removed from the conditioning chamber, or if the rocket motor temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall stop and the rocket motor shall be reconditioned. A reconditioning time of 2 minutes is required for every minute the rocket motor spent outside the specified temperature envelope. Reconditioning time shall not exceed 6 hours.

c. Excitation shall be applied through the aircraft seat attachment fittings unless otherwise noted in the contract (see 6.2.1). Accelerometers shall be placed midway between the attachment points to measure acceleration along the axis perpendicular to the axis through the attachment points of the rocket motor. Longitudinal vibration shall be measured at either end of the rocket motor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

d. Each test unit shall undergo random vibration testing per the "Aircraft with jet engines" vibration test of MIL-STD-2102. A successful candidate item shall meet the vibration criteria specified in MIL-STD-2102.

4.2 First article/production.

4.2.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: -65 ± 5°F
b. Hot: 165 ± 5°F

4.2.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 6 to 48 hours
b. Temperature cycling: 6 to 8 hours

4.2.3 Static firing. The PDRM shall be separated into its components (rocket section and mortar) and static fired individually.

4.2.3.1 PDRM rocket section. Each unit shall be temperature conditioned prior to testing. Each unit shall be conditioned for a minimum of 6 hours and not more than 48 hours. Within 5 minutes of removal from the conditioning chamber, the unit shall undergo ballistic testing. The following ballistic data shall be recorded: impulse, action time, thrust, ignition delay, initiation pressure, and thrust-time curve.

4.2.3.2 PDRM mortar. After each rocket section assembly has been tested in accordance with 4.2.3.1, the expended rocket section igniter cartridge (Drawing MBEU142790) shall be removed. A billet simulating the propellant grain mass and bridle assembly shall be inserted into the rocket tube. The rocket section shall be reassembled with a special test cartridge (Drawing EXP143934) with live primers only. The rocket motor shall then be reassembled. The total weight of the reassembled rocket section shall be 3.06 ± 0.10 lbm. The complete PDRM assembly shall then be temperature conditioned a minimum of 6 hours and not more than 48 hours. Within 5 minutes of removal from the conditioning chamber, each assembly shall undergo ballistic testing. Half of the units shall be electrically initiated and the other half ballistic gas initiated. If the units cannot be evenly divided, the extra unit shall be electrically initiated. The following data shall be recorded: ignition delay, velocity of motor, initiation pressure, and bridgewire resistance (regardless of test method) prior to test.

4.2.4 Radiographic inspection. All first article and production LAT units, at the test stages specified in 3.3.4, shall be radiographically inspected in accordance with MIL-A-85097.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

4.2.5 Resistance checks. The primary cartridge (Drawing MBEU142830) shall be resistance tested for insulation resistance between shorted pins-to-case and bridgewire circuit resistance after assembly into the rocket motor. The values obtained shall be within the tolerances stated in 3.3.5.

5. PACKAGING

(PDRMs shall be packaged per MIL-A-85097 and the applicable contract.)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rocket motor described herein is intended for use with SJU-17(V) Navy Aircrew Common Ejection Seat (NACES) as a means of deploying the personal parachute. It is used in the T-45A, F/A-18C/D, and F-14D aircraft.

6.1.1 General description. The Mark 122 Mod 0 Parachute Deployment Rocket Motor (PDRM) extracts the personal parachute from the parachute pack and enables the parachute to deploy and develop rapidly without becoming entangled with the seat. The PDRM is a sealed unit consisting of two main parts:

(a) a breech at its lower end containing an electrically initiated primary cartridge and a gas-operated secondary cartridge, and

(b) a rocket section with an integral gas-operated igniter cartridge.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Mk and Mod designation of rocket motor.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52-209-3 and -4 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

Paragraph no.	Data requirements title	Applicable DID no.
3.3.1	Acceptance Test Report	UDI-T-22741A

(Data items descriptions related to this specification, and identified in this section, will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions. The following ballistic performance definitions apply to the PDRM.

6.3.1 Impulse (lb-sec), rocket section. The area under the thrust-time curve taken over the action time.

6.3.2 Action time (sec), rocket section. The time interval from 50 percent of maximum thrust on the initial rise of the thrust-time curve to the corresponding 50 percent of maximum thrust on the declining portion of the curve.

6.3.3 Thrust (lb), rocket section. The maximum thrust measured on the thrust-time curve.

6.3.4 Ignition delay (sec), rocket section. The elapsed time from ignition signal (shear pin shear) to first indication of continuous thrust on the thrust-time curve.

6.3.5 Ignition delay (sec), mortar phase. The elapsed time from ignition signal (shear pin shear or fire current) to separation of motor (first movement).

6.3.6 Velocity of motor (ft/sec), mortar phase. The velocity of the motor measured over the first 6 inches of free travel.

6.3.7 Initiation pressure (psig), mortar phase. The applied pressure at shear pin shear, when applicable.

Preparing Activity:
Navy - OS
(Project 1377-NE82)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-85097/16	2. DOCUMENT DATE (YYMMDD) 5 December 1994

3. DOCUMENT TITLE
AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS) ROCKET MOTOR, PARACHUTE DEPLOYMENT, MARK 122 MOD 0

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office, Attn: DLSC-LM 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

DD Form 1426, OCT 89 *Previous editions are obsolete.*

Source: https://assist.dla.mil - Downloaded: 2026-10-23T10:24Z
Check the source to verify that this is the current version before use.